NATCHAYA LUANGPHAIRIN

PhD Candidate, Civil & Environmental Engineering, University of South Florida
Drinking Water Treatment, Access, Policy, and Sustainability (DW-TAPS) Research Group

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LinkedIn
GitHub
Google Scholar
Portfolio Website

PROFESSIONAL SUMMARY

Natchaya's career embodies interdisciplinary research and a commitment to advancing drinking water treatment, policy, and sustainable infrastructure. As a Ph.D. candidate in Civil Engineering, she brings more than five years of combined research and professional experience and over ten years of laboratory experience across academic, research, and professional settings. She integrates behavioral decision-making, participatory research, regulatory analysis, and physicochemical and biological treatment processes to address complex water-quality challenges. Her work spans the fate and transport of per- and polyfluoroalkyl substances (PFAS), 1-4, Dioxane, EPA regulatory-data analysis, potable reuse optimization, and biofiltration for emerging contaminant removal. She has authored multiple peer-reviewed publications, delivered over ten presentations at national conferences, and advanced the water sector through teaching, mentorship, technical workshops, and leadership in the American Water Works Association.

EDUCATION

Ph.D. in Civil Engineering, University of South Florida *Expected May 2029*
Dissertation: Determining Biofilter Design and Management Impacts on Biofilm Microbiome and Select Emerging Contaminant Degradation. Advisor: Dr. Katherine Alfredo, PhD, PE.

M.S. in Civil Engineering, University of South Florida *December 2024*
Focus: Water Policy, Risk Assessment, Distribution System Modeling

B.S. in Chemical Engineering, University of South Florida *May 2022*

A.S. in Biotechnology, Ivy Tech Community College – Bloomington *May 2017*
Recipient, Outstanding Student Award in Biotechnology, *Magna Cum Laude*

HONORS AND AWARDS

- **Design-Build Institute of America (DBIA) Scholarship** (2026)
- **Women of Water (WoW) Scholarship** (2026)
- **HSA Environmental Inc. Endowed Scholarship** (2025)
- **Florida Section American Water Works Association (FSAWWA) Roy W. Likins Scholarship** (2024)
- **JPB Environmental Health Fellowship (administered by Harvard University):** Using a Risk Cup approach to evaluate vulnerabilities associated with community water systems in the US. Served as a Graduate Researcher, with fellowship funding supporting portions of her RHI research. Principal Investigator: Dr. Katherine Alfredo. Total Project award: $200,000.
- **Project Lead, Association of State Drinking Water Administrators (ASDWA)** Nitrification Action Plan Project Funding (2022-2023). Total Project Funding: $12,072.
- **Research Assistant, FDACS Office of Agricultural Water Policy (FDACS Contract #28514)** (2024)
- **Phi Theta Kappa Honors Scholarship** (2018)

Between 2023 and 2026, Natchaya received **six best poster awards** across four national and regional water conferences, including one 1st-place, three 2nd-place, and two 3rd-place honors.

PROFESSIONAL AND ACADEMIC EXPERIENCE

Engineering Intern — Carollo Engineers *Jan 2025 – Present*
Water Group | Tampa, FL

- Led the development of a predictive tool to forecast influent PFAS concentrations for the City of Tampa, supporting proactive treatment planning and operational decision-making.
- Supported Suspended Ion Exchange pilot operations through laboratory and field analyses, data interpretation, and technical reporting; contributed to WRF Project 5380 on the co-treatment of dissolved organic matter and PFAS and the WRF Project 5418 proposal on optimizing PFAS treatment systems.
- Developed a qualitative PFAS risk and decision framework for wastewater-to-potable-reuse pilot projects by comparing source-water characteristics, treatment performance, and costs across XBAT, CBAT, and RBAT technologies.
- Designed a potable-reuse planning framework to optimize the treatment of total organic carbon, disinfection byproducts, contaminants of emerging concern, and PFAS in response to evolving federal regulatory requirements.

- Assessed the potential statewide effects of incorporating PFAS maximum contaminant levels into Chapter 62-550, Florida Administrative Code, by analyzing UIC, NPDES, and Group II Class V permit data and estimating the number of potentially affected Florida utilities.
- Built a performance dashboard for the Bee Ridge Wastewater Treatment Plant to monitor, troubleshoot, and optimize nutrient removal in its 5-stage Bardenpho process.
- Prepared a chemical-system startup plan for the City of Tampa's David L. Tippin Water Treatment Facility and supported a municipal bond report by compiling technical information and drafting sections on infrastructure needs and planned improvements.

Engineering Intern — Wright-Pierce *May 2025 – Dec 2025*

SE One Water Group | Tampa, FL

- Supported the design of municipal pump stations and evaluated force-main routing alternatives to inform infrastructure planning and system upgrades.
- Created field maps for the Pinellas Find & Fix sewer-rehabilitation project to guide manhole inspections, closed-circuit television inspections, and smoke testing.
- Mapped water-service-line networks by commissioner district and developed a system-asset inventory to support infrastructure assessment, prioritization, and planning.
- Evaluated emissions-characterization data and compared odor-control alternatives to support wastewater odor investigations and mitigation planning for the City of Orlando.
- Led technical knowledge-sharing sessions for engineering staff on biological treatment processes, water policy, federal environmental databases, and API-based data collection.

Graduate Research Assistant — USF Environmental & Water Resources Engineering Lab *May 2022 – Present*

Tampa, FL | Advancing water treatment techniques for micropollutant removal

- Redesigning biofilters and developing dynamic prediction models using deep-learning architectures (LSTM, Neural ODE, SINDy).
- Co-developed a novel biofilm-recovery method with the Southern Nevada Water Authority (SNWA) that improved the extraction of microbial material from granular activated carbon filters, supported molecular and flow-cytometric analyses, and resulted in a peer-reviewed publication.
- Developed an interactive R Shiny dashboard for the AWWA Inorganic Research Committee's Arsenic subgroup, enabling members to retrospectively evaluate arsenic removal technology's impacts and the 2001 MCL reduction across 50 EPA demonstration sites.
- Developed an interactive dashboard to track inorganic contaminant violations, identify geographic and temporal hotspots, and measure the time required for drinking water systems to return to compliance.
- Led a multi-level intervention analysis with the Association of State Drinking Water Administrators (ASDWA) evaluating statewide nitrification action plans and policy interventions on distribution-system nitrification, residuals, and DBPs.
- Conducted a comprehensive analysis of contaminant health risk using a novel Relative Health Indicator (RHI) metric to quantify potential health-risk reductions from PFAS regulation under varying standards, identifying significant regulatory opportunities.
- Developed a design and cost-estimation tool for Florida Department of Agriculture and Consumer Services (FDACS) published as a Best Management Practice Program (BMP) covering biochar-amended bioretention systems for nitrogen removal, supporting on-site nutrient management of nursery runoff.

Research Lab Manager — USF Research & Innovation *May 2022 – August 2026*

Tampa, FL | Oversee experimental lab supporting 30+ graduate students and 7 principal investigators

- Led laboratory orientations, emergency-response procedures, Environmental Health and Safety (EH&S) training, safety inspections, compressed-gas cylinder management, equipment maintenance, and chemical inventory control.
- Fostered a culture of safety that produced a significant reduction in lab violations, achieving zero incidents over four years.
- Developed laboratory safety standard operating procedures (SOPs) for flow cytometry, gas chromatography, and other analytical instruments to standardize safe operation, training, maintenance, and emergency response.

Research and Engineering Co-op — City of Tampa, Tippin Water Treatment Facility *Jan 2020 – April 2022*

Water Department | Tampa, FL

- Led treatment-optimization studies using jar testing, ATP testing, sieve analysis, chlorine-decay testing, and polymer evaluations to improve sludge thickening and dewatering.
- Investigated excessive filter head loss, identified media-replacement needs, and conducted pilot-scale biofiltration studies to optimize granular activated carbon particle size, filter runtime, and contaminant removal.
- Developed an Excel VBA-enabled tool adopted by the City to help field crews schedule, document, and quality-check hydrant flushing and inspection activities.
- Prepared monthly and annual aquifer storage and recovery reports for the FDEP and coordinated chlorine-maintenance events for the Water Department.
- Supported major water-reuse and research initiatives, including data collection for the Tampa Augmentation Project and research coordination with the SNWA for WRF#4912 project on harmful algal bloom monitoring and source-water control.

Undergraduate Student Researcher — C.L. Henderson Group *January 2020 – June 2020*
Chemical Engineering | Tampa, FL

- Modeled and simulated block copolymer phase separation using advanced molecular dynamics techniques and software such as MOE, VMD, HooMD.

Quality Control Intern — Catalent Pharma Solutions *June 2017 – May 2018*
Quality Control Analyst (QC) Stability | Bloomington, IN

- Expedited the formation of stability studies, revised and executed stability SOPs, produced stability reports,
- and helped QC fine tune Breakloose/Glideforce method and other analytical methods.
- Performed equipment calibration and Quality Control testing to determine adherence of product specifications.
- Applied analytical chemistry, microbiology, aseptic technique, spectrometry, and chromatography methods in accordance with GxP practices, FDA requirements, and safety, identity, strength, purity, and quality standards.
- Managed retained samples, chemical inventories, stability-room organization, laboratory supplies, and computer-based inventory records to maintain inspection readiness and uninterrupted laboratory operations.

Intern — National Center for Genetic Engineering and Biotechnology (BIOTECH) *June 2013 – August 2015*
Food Science Lab | Bangkok, Thailand

- Supported the research and development of a selective cultivation medium for Bifidobacterium growth.
- Delivered a video presentation used by the research center to educate volunteers on proper stool collecting.
- Prepared staining reagents, buffers, stock solutions; performed DNA extractions and gel electrophoresis analysis.
- Analyzed genomic fingerprinting patterns and maintained research data data using ImageLab, Gen Compar II.

PUBLICATIONS (ORCID: 0009-0002-2193-5449)

Natchaya is the author of **over 5 peer-reviewed publications, with several in preparation**, in journals including *Environmental Monitoring and Assessment*, *Water*, *AWWA Water Science*, and *Journal AWWA*, including an invited special column author spotlight.

8 Pickett, M., Kiger, A., **Luangphairin, N.**, Hart, V. (SUBMITTED/UNDER REVIEW, 2026) When Biology Isn't the Cause of Headloss: Ozone-Induced Microfloc Accumulation in Drinking Water Biofilters. AWWA Water Science.

7 **Luangphairin, N.**, Adhikari, M., Alfredo, K. (SUBMITTED/UNDER REVIEW). A Retrospective Cost and Performance Analysis of EPA's 50 Full-Scale Arsenic Pilots. AWWA Water Science.

6 Alfredo, K., & **Luangphairin, N**. Tracking regulated inorganic level violations (2026). Journal AWWA Water Quality Matters, 7(5), e70060. DOI: 10.1002/awwa.70060.

5 **Luangphairin, N.**, Roberson, A., Alfredo, K. (2025). Evaluating state nitrification action plans for controlling distribution system nitrification and MDBPs. AWWA Water Science, 118(3), e70036. DOI: 10.1002/aws2.70036.

4 Black, A., **Luangphairin, N.**, Papp, K., Gerrity, D., Dickenson, E., Alfredo, K. (2025). Enhancing biofilm recovery from carbon filters for molecular analysis and flow cytometry. AWWA Water Science, 7(4), e70029. DOI: 10.1002/aws2.70029.

3 Black, A., **Luangphairin, N.**, Alfredo, K. (2025). The impact of the fourth regulatory determination on vulnerable populations. Environmental Monitoring and Assessment, 197(323). PMID: 39992476.

2 Richardson, N., **Luangphairin, N.**, Bhattacharjee, A.S., Nachabe, M.H., Ergas, S.J. (2025). Nursery runoff treatment by novel biochar-amended bioretention systems. Water, 17(3), 330. DOI: 10.3390/w17030330.

1 Richardson, N., **Luangphairin, N.**, Cooper, R., Sharmin, N., Ergas, S.J., Nachabe, M., Bhattacharjee, A. (2024). Demonstration of Bioretention for Treatment of Agricultural Runoff. Published by the Florida Department of Agriculture and Consumer Services (FDACS), https://www.fdacs.gov/content/download/118071/file/FDACS%20_28514_Final_Report_revised.pdf.

Invited Spotlight. Natchaya has been featured in water-sector publications.

Luangphairin, N. AWWA Water Science Author Spotlight: Natchaya "Jade" Luangphairin. Journal AWWA (October 2026). https://doi.org/10.1002/awwa.70139

Luangphairin, N. AWWA active on college campuses. (September 5, 2025). https://www.awwa.org/AWWA-Articles/awwa-active-on-college-campuses/

Natchaya regularly contributes USF Student Chapter updates to the **quarterly FSAWWA Region IV newsletter**, highlighting the chapter's activities, student achievements, and water-sector engagement. https://www.fsawwa.org/page/320

Publications In Preparations. The manuscripts on this list are in draft form with some near completion.

Luangphairin, N., Alfredo, K. (*in preparation*). In Search of a Unified Model: The Evolution of Biofiltration Modeling for Drinking Water Treatment. Goal Journal: *Water Research*

Pickett, P., Kumar, P., Kiger, A., **Luangphairin, N.**, Thompson, K., Vicario, E. (*in preparation*). PFAS in Wastewater: Fate, Transport, and Treatment Insights from Potable Reuse Studies. Goal Journal: *Water Environment Research*

Luangphairin, N., Fee, S., Alfredo, K. (*in preparation*). Optimization of a GC–MS Method for the Detection and Quantification of 1,4-Dioxane in Drinking Water. Goal journal: *Water Research*

Luangphairin, N., Fee, S., Alfredo, K. (*in preparation*). Beyond the Plate Count: Tracking VBNCs and True Inactivation by Flow Cytometry Across a Ct Range. Goal journal: *Environmental Science & Technology*

Peer-Review Activities. Natchaya has served as a peer reviewer for the following manuscript on this list.

"Balancing Affordability and Cost Recovery: A Longitudinal Analysis of Water Bills," submitted for possible publication in AWWA Water Science in 2026.

"Six-year surveillance of fluoride and nitrate-nitrogen in municipal drinking water: Regulatory compliance and age-specific-upper-tail risk," submitted for possible publication in Scientific Reports in 2026.

RESEARCH PRESENTATIONS AND INVITED TALKS

Natchaya participated in **over 10 research presentations** at national and local conferences and has led workshops advancing database management practices and the application of statistical machine learning approaches in the water sector.

12 **Luangphairin, N.** Beyond the Plate Count: Tracking VBNCs and True Inactivation by Flow Cytometry Across a Ct Range. AWWA Water Quality Technology Conference (WQTC), Memphis, TN (2026, accepted)

11 **Luangphairin, N.** AWWA IMTR Workshop: From Data to Decision: A Framework for Policy Analysis Using Integrated EPA Datasets. AWWA Water Quality Technology Conference (WQTC), Memphis, TN (2026, accepted)

10 **Luangphairin, N.**, Alfredo, K. Achieving Arsenic Compliance: Lessons from EPA's 50 Demo Sites. AWWA Inorganic Committee, Teams (Jul 2026). Webcast.

9 **Luangphairin, N.**, Adhikari, M., Yazdani Arani, M., Alfredo, K. Arsenic Demo Sites: Where Are They Now? A Retrospective Cost and Performance Analysis of EPA's 50 Full-Scale Treatment Pilots. AWWA WQTC, Tacoma, WA (Nov 2025). Podium Presenter.

8 **Luangphairin, N**. Evaluating the Impact of State Nitrification Action Plans in Mitigating Distribution System Nitrification and MDBPs. FSAWWA Fall Conference, Kissimmee, FL (Dec 2024). Podium Presenter.

7 **Luangphairin, N**. Evaluating the Impact of State Nitrification Action Plans on System Nitrification, Disinfectant Residuals, and MDBP Risks. AWWA WQTC, Dallas, TX (Nov 2023). Podium Presenter.

6 **Luangphairin, N.**, Adhikari, M., Alfredo, K. Arsenic Demo Sites: Where Are They Now? SWFWRC/AWRA, Fort Myers, FL (Jan 2026). Best Poster, 1st Place.

5 **Luangphairin, N.**, Adhikari, M., Alfredo, K. Arsenic Demo Sites: Where Are They Now? FSAWWA Conference, Orlando, FL (Nov–Dec 2025). Best Poster, 2nd Place.

4 Black, A., **Luangphairin, N.**, Papp, K., Gerrity, D., Dickenson, E., Alfredo, K. Biofilm Removal from Granular Media Filters for Microbial Analysis. AWWA WQTC, Tacoma, WA (Nov 2025). Best Poster, 3rd Place.

3 **Luangphairin, N.**, Roberson, A., Alfredo, K. Evaluating the Impact of State Nitrification Action Plans on Distribution System Nitrification and MDBP Production. FWRC Conference, Kissimmee, FL (Apr 2024). Best Poster, 2nd Place.

2 **Luangphairin, N.**, Roberson, A., Alfredo, K. Evaluating the Impact of Nitrification Action Plans on System Nitrification, Residuals, and MDBP Risks. FSAWWA Conference, Championsgate, FL (Dec 2023). Best Poster, 2nd Place.

1 **Luangphairin, N.**, Roberson, A., Alfredo, K. Evaluating the Impact of Nitrification Action Plans on System Nitrification and MDBP Production. SWFWRC/AWRA, Fort Myers, FL (Feb 2023). Best Poster, 3rd Place.

PROFESSIONAL AFFILIATIONS & SERVICE

At USF Natchaya served as the laboratory manager for the Environmental and Water Resources Engineering laboratory. Her services have improved lab safety and compliance (reduced laboratory violations and reported injuries to zero) and reflect her deep commitment to improving engineering training and career readiness at her institution. Additionally, she is invited to peer review papers and served on national committees as active members and leading local Chapter of the American Water Works Association, and mentors students and water professionals. She also contributes to Operation WETVET, which recognizes and supports veteran members of the Florida Section AWWA, and Operation FISHHOOK, which promotes water-sector career pathways for veterans. Her service and mentorship demonstrate her commitment to engagement and outreach in the water sector.

President, American Water Works Association (AWWA)
USF Student Chapter, Region IV (Oct 2023 – Present)

Lead monthly board meetings, manage the organization's presence, and coordinate outreach, volunteer beach/park cleanups, social, and networking events that connect students, faculty, and water-industry professionals, strengthen member engagement, and advance AWWA's vision of "*A Better World Through Better Water*".

Marketing Chair, AWWA Information Management & Technology Research (IMTR) Committee
Committee, National, (Dec 2025 – Present)

AWWA IMTR GitHub & LinkedIn Page Administrator.

Active Member/Event Co-organizer, Florida Water Environment Association (FWEA),
USF Student Chapter, West Coast Chapter (Oct 2023 – Present)

Promote student engagement, professional visibility, and water-environment outreach.

Active Member of the Society of Women Engineers (SWE) *(June 2026– Present)*

Science Fair Judge
Volunteer Judge

Hillsborough Regional STEM Fair (*Feb 2026*)
Water Tower Model design/build competition (*Feb 2026*)
USF Florida Invention Convention Competition at Largo STEM Fair (*Jan 2026*)
Roland Park K–8 STEM Fair (*Nov/Dec 2025/Present*)

Great American Teach-In
Teach-In Volunteer

Led drinking water treatment demonstrations for 1st–3rd graders at Walton Academy Charter School and Roland Park Magnet School to promote STEM education (*Nov 2024/2025*)

Graduate Mentor, **National Science Foundation Research Experiences for Undergraduates (NSF REU)**
NSF Summer REU Mentor (Summer 2022/2024, Present)

TEACHING EXPERIENCE

Natchaya has taught undergraduate- and graduate-level lessons covering analytical methods for water-quality testing, navigation and analysis of federal environmental databases, introductory R programming and statistical analysis, and advanced multilevel modeling. The following section details the courses and workshops in which she has served as an instructor or guest lecturer at USF and national conferences, teaching utility operators, researchers, engineers, consultants, and students.

- **Workshop Presenter:** "*From Data to Decision: A Framework for Policy Analysis Using Integrated EPA Datasets*," AWWA IMTR Committee Workshop. Accepted for presentation at the 2026 WQTC Conference, Memphis, TN.
- **Guest Teaching Assistant:** ENV2061 Engineering Sustainable & Healthy Environments—*Tragedy or Cooperation of the Commons Simulation (Fall 2024/2026)*
- **Guest Lecturer:** ENV6666 Aquatic Chemistry—*Safe Drinking Water Act, USEPA Databases, and Regulations (Spring 2022/2023, Fall 2024/2025/2026)*
- **Guest Teaching Assistant:** ENV6666 Aquatic Chemistry—*Aquatic Chemistry Titration Lab (Fall 2024/2025)*
- **Graduate Teaching Assistant:** ENV4004L Environmental Engineering Lab—*Lab Techniques (Sept–Dec 2023)*
- **Instructor:** DW-TAPS—*Intro to R for Statistical Analysis and Multi-level modeling Workshop Series (Jan 2024)*

Mentorship. Natchaya currently mentors one undergraduate researcher and has served as a **graduate mentor for four NSF REU cohorts** at USF, written recommendation letters and guiding undergraduate students from institutions across the United States. Her mentees have gone on to present their research at academic symposiums, including:

- **Sarah Fee**, University of South Florida, Environmental Engineering Student (*USF REU, Fall 2024-present*)
  Project title: *The Invisible Threat: Traditional culture-based monitoring misses the VBNC state*
- **Marie Emmanuelle Compere**, College of Central Florida, Biology Student (*NSF REU, Summer 2024*)
  Project title: *Evaluating Seasonal Impacts on Microbial Growth and Contaminant Removal in a Drinking Water Biological Filter*
- **Toheedat Aderemi Bakare**, University of North Carolina-Chapel Hill, Environmental Health Science Student (*NSF REU, Summer 2022*)
  Project title: *The Use of Biofiltration in Degradation of Salicylic Acid and nitrate Pollution*
- **Sky Embray**, Trinity College, Environmental Science, Public Policy & Law Student (*NSF REU, Summer 2022*)
  Project title: *The Efficiency and Range of Biofiltration in Drinking Water Treatment*

PROFESSIONAL SKILLS AND CERTIFICATIONS

Skills: R (Shiny), Python, Tableau, Power BI, PostgreSQL, ArcGIS, BioWin, GPS-X, Aspen, COMSOL Multiphysics, SCADA, MATLAB & Simulink, Adobe Creative Suite (Photoshop, Illustrator), Microsoft Office Suite.

Certifications: Engineer Intern (E.I.)—Florida Board of Professional Engineers, Environmental Sustainability Professional (ENV SP), Six Sigma, ICPSR Statistics and Quantitative Methods Certification, CPR/AED/First Aid, Chemical Engineering Education (SAChE), Responsible Conduct of Research for Engineering (CITI Program), OSHA10.

Core Technical Competencies: Drinking water treatment and biofiltration; PFAS and emerging contaminant research; potable reuse; GC–MS and chromatography methods; flow cytometry and microbiological methods; water-quality analysis; process modeling and optimization using BioWin, GPS-X, and Aspen; statistical and geospatial analysis using R and ArcGIS; EPA regulatory-data analysis; web development.